

Tan Yong Chin

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Personal Summary

Computer Science undergraduate with strong foundations in Python programming, software engineering, and data processing.

I have experience supporting application development, automating workflows, and working with structured data through coursework and projects.

My prior diploma in Business Studies has given me a strong understanding of financial markets and business processes, which helps me translate operational requirements into technical solutions.

I am excited to contribute to technology solutions that support complex financial workflows, and I enjoy collaborating with cross functional teams to design, test, and maintain reliable systems.

Skills

Languages Python, C++, C#, SQL
Frameworks OpenGL
Tools Git, GitHub

Knowledge ML Fundamentals, Application Development, Workflow Automation

Education

Singapore Institute of Technology & DigiPen Institute of Technology
Bachelor of Science in Computer Science in Real-Time Interactive Simulation
GPA: 4.53

Aug 2023 – Present
Expected Graduation: Apr 2027

Ngee Ann Polytechnic
Diploma in Business Studies

Apr 2017 – Apr 2020

Projects

Real-Time Simulation Engine [\[Link\]](#)

Graphics Lead — C++, C#, OpenGL

A custom real-time simulation and rendering framework built to support a 3D interactive application.

- Designed a modular GPU-optimised rendering architecture with advanced features such as clustered forward+ lighting, visibility buffer pipeline, instancing, bind-less textures, and pipeline caching to support scalable performance.
- Improved runtime performance by profiling GPU and CPU workloads, removing bottlenecks, and reducing expensive driver calls to maintain high and stable frame rates.
- Collaborated closely with designers and engineers to translate creative requirements into technical specs, ensuring the graphics system enhanced both visual quality and user experience.

Geo-Regulation Compliance Detection System [\[Link\]](#)

Full-Stack Developer — Python, Streamlit, Gemini API, Ethereum, FAISS

A tool that uses LLMs to flag features for geo-compliance, generating detailed audit logs whose integrity is immutably proven via blockchain verification.

- Architected a full-stack Streamlit app with a multi-agent RAG pipeline and Gemini API integration, delivering accurate, source-cited compliance analysis.
- Designed and implemented an Ethereum-based blockchain verification system that cryptographically secured audit logs, creating a tamper-proof and legally verifiable chain of custody.
- Led a team through iterative development cycles, coordinating front-end, back-end, and AI tracks to successfully deliver a fully functional prototype on schedule.

Experience

DBS Bank
Human Resources Intern

Mar 2019 – Aug 2019

- Supported process automation and improved accuracy in employee data handling.
- Collaborated with cross-functional teams to ensure smooth onboarding and internal operations.
- Applied analytical thinking to investigate discrepancies and propose improvements.